

## Absolute Value Inequalities: Special Cases and Interval Notation - Practice

Algebra II Honors | Unit 2 | Lesson 8 Practice | TEK A2.6.F

**Objective:** I can write absolute value inequality solutions in interval notation, and explain why the same isolated expression can produce either no solution or all real numbers.

### Section 1 — Recall

1. Write  $|x| \geq 11$  in interval notation.

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2. Write  $|x| < 8$  in interval notation.

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### Section 2 — Apply

3. Solve  $2|3x - 2| + 5 < 29$ , and write the answer in interval notation.

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4. Solve  $|4x + 3| - 7 \geq 14$ , and write the answer in interval notation.

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### Section 3 — Challenge

5. Solve  $3 + |x - 5| \geq 1$ .

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6. Solve  $3 + |x - 5| < 1$ .

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### Section 4 — Explain It

7. Items 5 and 6 both isolate to  $|x - 5|$  compared to  $-2$ , yet one has EVERY real number as a solution and the other has NONE. Explain why flipping the inequality symbol produces such opposite results.

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